

Initialize Stage—Casual Format

Main Success Scenario

1. With the System running, the Flight Director enters the Mission Parameters File (See Use Case 1 Initialize Launch)
2. System Initializes each Engine in the Stage (See *Use Case 1.1.1.1 Initialize Engine*)
3. System Initializes the Fuel System in the Stage (See *Use Case 1.1.1.2 Initialize Fuel System*)
4. System finds the Stage Parameters File name and location
5. System goes to the Stage Parameters File location
6. System opens the Stage Parameters File
7. System reads the Stage Parameters File
8. System initializes the Stage Data with the appropriate Stage Data read in from the Stage Parameters file
9. System returns the Initialization Status to the Flight Director

Alternative Scenarios

- 2a If the System receives an Initialization Error from the Engines, then the System returns the Error to the Flight Director (See *Use Case 1.1.1.1 Initialize Engine*)
- 3a If the System receives an Initialization Error from the Fuel System, then the System returns the Error to the Flight Director (See *Use Case 1.1.1.2 Initialize Fuel System*)
- 4a,5a If the System cannot find the location where the Stage Parameters File is located, then the System returns an Error to the Flight Director: indicating a non-legitimate file path.
- 6a If the System cannot open the Stage Parameters file, then the System returns an Error to the Flight Director: indicating unable to open the Stage Parameters file.
- 7a If the System cannot read the Stage Parameters File, then the System returns an Error to the Flight Director: indicating unable to read the Stage Parameters file.

Notes/Constraints

- 9a The System indirectly alerts the Stake Holders via an upward cascade of alerts/queries (See *Use Cases 1.1.1.1 Initialize Engine, 1.1.1.2 Initialize Fuel System*)